

6-1-1: Examining the 802.11 Standard

At the end of this episode, I will be able to:

1. Identify and explain the IEEE 802.11 standards, given a scenario.

Learner Objective: *Identify and explain the IEEE 802.11 standards, given a scenario.*

Description: In this episode, the learner will examine the IEEE's 802.11 family of wireless LAN (WLAN) standard. We will examine the development of the standard, including the frequencies, channels, speeds of these standards, and more.

- What are the 802.11 standards?
 - o The IEEE's working group for wireless local area networks (WLAN)
- What are the 802.11 standards to be aware of?
 - o 802.11a, .11b, .11g, .11n, .11ac, .11ax, .11be (in development)
- What are a few of the characteristics of the various 802.11 standards are important factors?
 - o 802.11a (Wi-Fi 2):
 - ♣ Speed - Up to 54 Mbps
 - ♣ Frequency - 5 GHz
 - ♣ Range - Shorter range due to higher frequency
 - ♣ Interference - Less interference
 - ♣ Wi-Fi Alliance Name - Wi-Fi 2 (unofficially)
 - o 802.11b (Wi-Fi 1):
 - ♣ Speed - Up to 11 Mbps
 - ♣ Frequency - 2.4 GHz
 - ♣ Range - Good range due to lower frequency
 - ♣ Interference - More interference (many devices use 2.4 GHz)
 - ♣ Wi-Fi Alliance Name - Wi-Fi 1 (unofficially)
 - o 802.11g (Wi-Fi 3):
 - ♣ Speed - Up to 54 Mbps
 - ♣ Frequency - 2.4 GHz
 - ♣ Range - Good range, similar to 802.11b
 - ♣ Interference - More interference (2.4 GHz is commonly used)
 - ♣ Wi-Fi Alliance Name - Wi-Fi 3 (unofficially)
 - o 802.11n (Wi-Fi 4):
 - ♣ Speed - Up to 600 Mbps
 - ♣ Frequency - 2.4 GHz and 5 GHz
 - ♣ Range - Better than 802.11a/g, option for 2.4 GHz improves range
 - ♣ Interference - The 2.4 GHz band has more interference, less on the 5 GHz
 - ♣ Wi-Fi Alliance Name - Wi-Fi 4
 - ♣ MIMO - Supported (up to 4x4)
 - o 802.11ac (Wi-Fi 5):
 - ♣ Speed - Up to 6.9 Gbps
 - ♣ Frequency - 5 GHz
 - ♣ Range - Shorter than 2.4 GHz networks, better with beamforming
 - ♣ Interference - Less crowded, minimal interference
 - ♣ Wi-Fi Alliance Name - Wi-Fi 5
 - ♣ MIMO - Supported (up to 8x8)
 - ♣ MU-MIMO - Supported
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- 802.11ax (Wi-Fi 6):

- ♣ Speed - Up to 9.6 Gbps
- ♣ Frequency - 2.4 GHz and 5 GHz
- ♣ Range - Improved over Wi-Fi 5, especially in the 2.4 GHz band
- ♣ Interference - Technologies like OFDMA reduce interference
- ♣ Wi-Fi Alliance Name - Wi-Fi 6
- ♣ MIMO - Supported (up to 8x8)
- ♣ MU-MIMO - Supported

o 802.11be (Wi-Fi 7):

- ♣ Speed - Expected to exceed 30 Gbps
- ♣ Frequency - 2.4 GHz, 5 GHz, and 6 GHz
- ♣ Range - Anticipated to offer enhanced range with new technologies
- ♣ Interference - Expected to be managed with advanced techniques
- ♣ Wi-Fi Alliance Name - Wi-Fi 7 (anticipated)
- ♣ MIMO - Supported (expected to support higher than 8x8)
- ♣ MU-MIMO - Supported

- What is MIMO (Multiple Input Multiple Output)? and MU-MIMO (Multi-User MIMO)?
 - o MIMO - A technology using multiple antennas at both the transmitter and receiver.
 - o MIMO - Enhances communication performance, data rates, and network capacity by using multiple signal paths.
 - o MU-MIMO - An advancement of MIMO technology.
 - o MU-MIMO - Allows multiple users to simultaneously access the same network.
 - o MU-MIMO - Each user has their own dedicated set of multiple antennas, improving network efficiency and throughput.
 - Why are channels important in WLANs?
 - o Each frequency that is used has specific channels and channel numbers
 - o 2.4 GHz - uses 11 channels in North America
 - o 2.4 GHz - has three non-overlapping channels (1, 3, 11)
 - o 5 GHz - all channels are non-overlapping
 - o 5 GHz - bands are labeled UNII and some channels support dynamic frequency selection to allow wireless communication in the same channels as radar operates.
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- Additional Resources:
 - o If applicable